

Problem 6

Problem 6

Problem 1 Two functions, h and g , are given

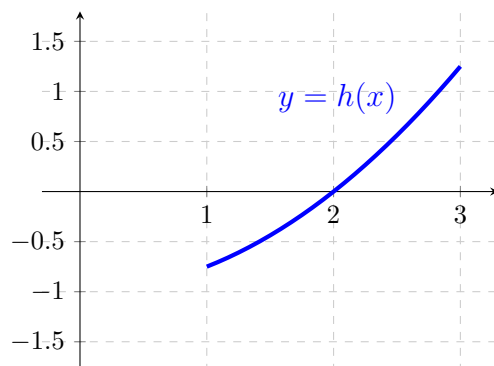
$$h(x) = \frac{x^2 - 4}{4}, \quad 1 < x < 3$$

$$g(x) = x - 2, \quad 1 < x < 3$$

The graph of the function h is given in the figure below. Let f be a function defined on the interval $(1, 3)$ that satisfies the following inequalities

$$g(x) \leq f(x) \leq h(x), \quad 1 < x < 3$$

- (a) In the figure below, sketch and label the graph of g and a possible graph of f . (All three functions have a common domain $(1, 3)$.)



- (b) Evaluate the limit, or state that it does not exist. Justify your answer.

(i) $\lim_{x \rightarrow 2} f(x)$

(ii) $\lim_{x \rightarrow 2} \frac{f(x) + 2}{x - 1}$

(iii) $\lim_{x \rightarrow 2} g(1 + e^{f(x)})$